A unital Banach algebra which is not a Calkin algebra

Draft proof for verification

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Abstract

We construct a unital complex Banach algebra which is not isomorphic, as a Banach algebra, to $\mathcal{B}(X)/\mathcal{K}(X)$ for any Banach space X. Here "isomorphic as Banach algebras" means a bounded complex-linear algebra isomorphism; by the open mapping theorem the inverse is then bounded. The proof is topological: it uses density, quotient norms, and closed ideals. It does not address purely algebraic discontinuous isomorphisms after forgetting the Banach-space norms.

1 Conventions and notation

All Banach spaces and Banach algebras are complex. For a Banach space X, write

$$\mathcal{B}(X) = \text{the Banach algebra of bounded operators } X \to X,$$

$$\mathcal{K}(X) = \text{the closed ideal of compact operators on } X,$$

and

$$\mathcal{C}(X) = \mathcal{B}(X)/\mathcal{K}(X).$$

The density character of a metric space Y is denoted by $\operatorname{dens} Y$. A Banach algebra is called *topologically simple* if it has no non-zero proper closed two-sided ideals.

Throughout, a Banach-algebra isomorphism means a bounded complex-linear multiplicative bijection. Such a map is automatically a homeomorphism by the open mapping theorem. Between unital algebras, surjectivity also forces it to preserve the identity.

The proof uses two standard external facts.

(i) Schauder's theorem: if K is compact, then K^* is compact.

(ii) The Josefson–Nissenzweig theorem: if X is infinite-dimensional, then there is a sequence $(f_n) \subseteq X^*$ such that $\|f_n\| = 1$ for all n, and $f_n(x) \to 0$ for every $x \in X$.

Everything else is proved below.

2 The algebra

Let

$$\lambda = \beth_\omega = \sup_{n<\omega} \beth_n.$$

Then λ is a strong-limit cardinal and $\lambda > 2^{\aleph_0}$. Let Γ be a set with $|\Gamma| = \lambda$, and let

$$W = \Gamma^{<\omega}$$

be the set of all finite words over Γ, including the empty word $\varnothing$. Then $|W| = \lambda$. Put

$$E = c_0(W).$$

For $w \in W$, let e_w denote the usual unit vector in $c_0(W)$.

For each $\alpha \in \Gamma$, define $S_\alpha, T_\alpha \in \mathcal{B}(E)$ on the unit-vector basis by

$$S_\alpha e_w = e_{\alpha^\frown w},$$

and

$$T_\alpha e_{\beta^\frown w} = \begin{cases} e_w, & \beta = \alpha, \\ 0, & \beta \neq \alpha, \end{cases} \qquad T_\alpha e_\varnothing = 0.$$

Here $\alpha^\frown w$ denotes concatenation. These formulas define contractions on $c_0(W)$, and in fact

$$\|S_\alpha\| = \|T_\alpha\| = 1.$$

They also satisfy

$$T_\alpha S_\beta = \delta_{\alpha\beta} I_E.$$

Moreover, if $F \subseteq \Gamma$ is finite and $(a_\alpha)_{\alpha \in F}$ is a scalar family, then

$$\left\| \sum_{\alpha \in F} a_\alpha S_\alpha \right\| = \max_{\alpha \in F} |a_\alpha|. \tag{1}$$

Indeed, the ranges $S_\alpha E$ sit on pairwise disjoint cylinders $\alpha^\frown W$.

Let B_λ be the closed unital subalgebra of $\mathcal{B}(E)$ generated by

$$I_E, \qquad S_\alpha, \qquad T_\alpha \quad (\alpha \in \Gamma).$$

Choose a maximal proper closed two-sided ideal

$$M \lhd B_\lambda,$$

and put

$$A_\lambda = B_\lambda / M.$$

Such an ideal exists by Zorn's lemma. Indeed, if (M_i) is a chain of proper closed two-sided ideals, then $\bigcup_i M_i$ is a two-sided ideal, and its closure is still proper: otherwise some element of $\bigcup_i M_i$ would be within distance strictly less than 1 of I_E, hence would be invertible, forcing one of the M_i's to be all of B_λ.

Thus A_λ is a unital topologically simple Banach algebra.

3 The density of A_λ

Lemma 3.1. $\operatorname{dens} A_\lambda = \lambda$.

Proof. First, B_λ is generated as a closed algebra by λ many elements. Noncommutative polynomials with coefficients in $\mathbb{Q} + i\mathbb{Q}$ and variables from a finite subset of the generators form a dense subset of cardinality at most λ. Hence

$$\operatorname{dens} B_\lambda \leq \lambda,$$

and therefore

$$\operatorname{dens} A_\lambda \leq \lambda.$$

For the reverse inequality, write

$$\sigma_\alpha = S_\alpha + M, \qquad \tau_\alpha = T_\alpha + M$$

inside A_λ. If $\alpha \neq \beta$, then

$$\tau_\alpha(\sigma_\alpha - \sigma_\beta) = 1_{A_\lambda}.$$

The quotient norm satisfies $\|\tau_\alpha\| \le \|T_\alpha\| = 1$. Also $\|1_{A_\lambda}\| = 1$: if $\operatorname{dist}(I_E, M) < 1$, then some $m \in M$ satisfies $\|I_E - m\| < 1$, so m is invertible, and hence $M = B_\lambda$, contradiction. Therefore

$$1 = \|1_{A_\lambda}\| = \|\tau_\alpha(\sigma_\alpha - \sigma_\beta)\| \le \|\tau_\alpha\| \, \|\sigma_\alpha - \sigma_\beta\| \le \|\sigma_\alpha - \sigma_\beta\|.$$

Thus $\{\sigma_\alpha : \alpha \in \Gamma\}$ is a 1-separated subset of A_λ of cardinality λ. Every dense subset must meet the pairwise disjoint balls of radius $1/3$ around these points, so $\operatorname{dens} A_\lambda \ge \lambda$. Hence $\operatorname{dens} A_\lambda = \lambda$. $\square$

4 A density trap for Calkin algebras

Lemma 4.1. *Let X be a Banach space. If $\operatorname{dens} X = \kappa$ is infinite, then*

$$\operatorname{dens} \mathcal{C}(X) \le 2^\kappa.$$

Consequently, if $\mathcal{C}(X)$ is Banach-algebra isomorphic to A_λ, then

$$\operatorname{dens} X \ge \lambda > 2^{\aleph_0}.$$

Proof. A metric space of density κ has cardinality at most $\kappa^{\aleph_0}$. Thus

$$|X| \le \kappa^{\aleph_0}.$$

Every operator in $\mathcal{B}(X)$ is determined by its values on a fixed dense subset of cardinality κ. Therefore

$$|\mathcal{B}(X)| \le |X|^\kappa \le (\kappa^{\aleph_0})^\kappa = \kappa^\kappa = 2^\kappa.$$

Thus $\operatorname{dens} \mathcal{C}(X) \le |\mathcal{C}(X)| \le |\mathcal{B}(X)| \le 2^\kappa$.

Now suppose $\mathcal{C}(X)$ is Banach-algebra isomorphic to A_λ. Since A_λ has density λ, this already excludes finite-dimensional X; hence $\kappa = \operatorname{dens} X$ is infinite. Banach-algebra isomorphisms are homeomorphisms, so

$$\operatorname{dens} \mathcal{C}(X) = \operatorname{dens} A_\lambda = \lambda.$$

If $\kappa < \lambda$, then, since $\lambda = \beth_\omega$ is strong limit, $2^\kappa < \lambda$, contradiction. Hence $\kappa \ge \lambda$. In particular, $\operatorname{dens} X > 2^{\aleph_0}$. $\square$

5 Killing countably many compact errors

Lemma 5.1. *Let X be a Banach space with $\operatorname{dens} X > 2^{\aleph_0}$. Let (K_n) be a countable family of compact operators on X, and let (ε_n) be a sequence of positive real numbers. Then there is $x \in X$ such that*

$$\|x\| = 1 \qquad \text{and} \qquad \|K_n x\| \le \varepsilon_n \quad (n \in \mathbb{N}).$$

Proof. Fix a compact operator $K \in \mathcal{K}(X)$ and $\varepsilon > 0$. By Schauder's theorem, $K^* : X^* \to X^*$ is compact. Hence $K^*(B_{X^*})$ is norm-totally bounded. Choose $g_1, \ldots, g_m \in B_{X^*}$ such that for every $f \in B_{X^*}$ there is $r \in \{1, \ldots, m\}$ with

$$\|K^* f - K^* g_r\| < \varepsilon.$$

Set

$$Y = \bigcap_{r=1}^{m} \ker(K^* g_r).$$

Then Y has finite codimension. If $y \in Y$, then

$$\|Ky\| = \sup_{f \in B_{X^*}} |f(Ky)| = \sup_{f \in B_{X^*}} |K^* f(y)| \leq \varepsilon \|y\|.$$

Thus for each K_n there is a finite-codimensional closed subspace $Y_n \subseteq X$ such that

$$\|K_n|_{Y_n}\| \leq \varepsilon_n.$$

Let

$$Y_\infty = \bigcap_{n=1}^{\infty} Y_n.$$

If $Y_\infty = \{0\}$, then the map

$$X \longrightarrow \prod_{n=1}^{\infty} X/Y_n, \qquad x \longmapsto (x + Y_n)_n$$

is injective. Each quotient X/Y_n is finite-dimensional over $\mathbb{C}$, so the product has cardinality at most

$$(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}.$$

This would imply $|X| \leq 2^{\aleph_0}$, contradicting $\operatorname{dens} X > 2^{\aleph_0}$. Hence $Y_\infty \neq \{0\}$. Choose $0 \neq x \in Y_\infty$, and normalize it. Then $\|x\| = 1$ and $\|K_n x\| \leq \varepsilon_n$ for every n. $\square$

6 The column system forces a copy of c_0

Lemma 6.1. *Let X be a Banach space with $\operatorname{dens} X > 2^{\aleph_0}$. Suppose there are elements $(s_j)_{j\geq 1}$, $(t_j)_{j\geq 1}$ in $\mathcal{C}(X)$ such that*

(a) $t_i s_j = \delta_{ij} 1_{\mathcal{C}(X)}$ for all i, j;

(b) there is $C < \infty$ such that, for every finitely supported scalar family (a_j),

$$\left\| \sum_j a_j s_j \right\| \leq C \max_j |a_j|;$$

(c) $\sup_j \|t_j\| < \infty$.

Then X contains an isomorphic copy of c_0.

Proof. Choose arbitrary lifts $S_j \in \mathcal{B}(X)$ of the s_j's. Since $\sup_j \|t_j\| < \infty$, choose lifts $T_j \in \mathcal{B}(X)$ of the t_j's with $\|T_j\| \leq \|t_j\| + 1$. Then

$$D := \max \left\{ 1, \sup_j \|T_j\| \right\} < \infty.$$

For all i, j, define

$$R_{ij} = T_i S_j - \delta_{ij} I_X.$$

By (a), $R_{ij} \in \mathcal{K}(X)$.

Let $\mathbb{Q}(i) = \mathbb{Q} + i\mathbb{Q}$. For every non-zero rational finitely supported scalar family

$$a = (a_j) \in c_{00}(\mathbb{Q}(i)), \qquad m(a) := \max_j |a_j|,$$

choose $K_a \in \mathcal{K}(X)$ such that

$$\left\| \sum_j a_j S_j + K_a \right\| \leq (C+1)m(a). \tag{2}$$

This is possible because

$$\left\| \sum_j a_j s_j \right\| \leq Cm(a)$$

and the norm in the quotient $\mathcal{B}(X)/\mathcal{K}(X)$ is an infimum over compact perturbations.

The family

$$\{R_{ij} : i, j \in \mathbb{N}\} \cup \{K_a : 0 \neq a \in c_{00}(\mathbb{Q}(i))\}$$

is countable. Apply Lemma 5.1 to this family with the following tolerances:

$$\|R_{ij}x\| \leq 2^{-j-1} \quad (i, j \in \mathbb{N}),$$

and

$$\|K_a x\| \leq m(a) \quad (0 \neq a \in c_{00}(\mathbb{Q}(i))).$$

We obtain a unit vector $x \in X$ satisfying all these inequalities.

Set

$$x_j = S_j x \qquad (j \geq 1).$$

We claim that (x_j) is equivalent to the canonical basis of c_0.

First let $0 \neq a = (a_j) \in c_{00}(\mathbb{Q}(i))$. Using (2),

$$\begin{aligned}
\left\| \sum_j a_j x_j \right\| &= \left\| \sum_j a_j S_j x \right\| \\
&\leq \left\| \left(\sum_j a_j S_j + K_a \right) x \right\| + \|K_a x\| \\
&\leq (C+1)m(a) + m(a) = (C+2)m(a).
\end{aligned}$$

For the lower estimate, choose i such that $|a_i| = m(a)$. Then

$$T_i \left(\sum_j a_j x_j \right) = a_i x + \sum_j a_j R_{ij} x.$$

Hence, since $\|x\| = 1$,

$$\begin{aligned}
\left\| T_i \left(\sum_j a_j x_j \right) \right\| &\geq |a_i| - \sum_j |a_j| \, \|R_{ij}x\| \\
&\geq m(a) - m(a) \sum_{j=1}^{\infty} 2^{-j-1} = \frac{1}{2} m(a).
\end{aligned}$$

Since $\|T_i\| \leq D$,

$$\left\| \sum_j a_j x_j \right\| \geq \frac{1}{2D} m(a).$$

Thus, for rational finitely supported scalar families,

$$\frac{1}{2D} \max_j |a_j| \leq \left\| \sum_j a_j x_j \right\| \leq (C+2) \max_j |a_j|.$$

By density of $c_{00}(\mathbb{Q}(i))$ in each finite-dimensional coordinate subspace, the same estimates hold for all finitely supported complex scalar families. Therefore the linear map

$$c_{00} \longrightarrow X, \qquad e_j \longmapsto x_j$$

extends to an isomorphic embedding $c_0 \hookrightarrow X$. $\square$

7 If a nonseparable space contains c_0, its Calkin algebra is not topologically simple

Lemma 7.1. *Let X be a nonseparable Banach space. If X contains an isomorphic copy of c_0, then $\mathcal{C}(X)$ has a non-zero proper closed two-sided ideal. In particular, $\mathcal{C}(X)$ is not topologically simple.*

Proof. Let

$$i : c_0 \hookrightarrow X$$

be an isomorphic embedding. By the Josefson–Nissenzweig theorem, choose $(f_n) \subseteq X^*$ such that

$$\|f_n\| = 1 \quad (n \geq 1), \qquad f_n(x) \to 0 \quad (x \in X).$$

Define

$$J : X \to c_0, \qquad Jx = (f_n(x))_{n=1}^{\infty}.$$

This is a bounded operator, and it maps into c_0 because $f_n(x) \to 0$ for every $x \in X$.

The operator J is not compact. Indeed, compact subsets $L \subseteq c_0$ have uniformly vanishing tails:

$$\lim_{n\to\infty} \sup_{y\in L} |y_n| = 0.$$

To see this, take a finite ε-net for L and use that each net point lies in c_0. But for $J(B_X)$,

$$\sup_{\|x\|\leq 1} |(Jx)_n| = \sup_{\|x\|\leq 1} |f_n(x)| = \|f_n\| = 1$$

for every n. Hence $J(B_X)$ is not relatively compact.

Therefore $iJ : X \to X$ has separable range and is noncompact. The range is contained in $i(c_0)$. If iJ were compact, then applying the bounded inverse $i^{-1} : i(c_0) \to c_0$ to the relatively compact set $(iJ)(B_X)$ would imply that $J(B_X)$ is relatively compact in c_0, contradiction.

Now let

$$\mathcal{S}(X) = \{T \in \mathcal{B}(X) : \overline{T[X]} \text{ is separable}\}.$$

This is a closed two-sided ideal of $\mathcal{B}(X)$. For completeness, here is the verification. It is an ideal because composition with bounded operators preserves separability of the closed range, and sums of two separable-range operators again have separable range. It is norm-closed because if $T_n \to T$ and each T_n has separable range, then $\overline{T[X]}$ is contained in the closed linear span of the countable union of separable subspaces $\overline{T_n[X]}$.

Every compact operator has separable range closure, because $K[X] = \bigcup_{n\geq 1} nK[B_X]$ and each $\overline{K[B_X]}$ is compact metrizable, hence separable. Thus

$$\mathcal{K}(X) \subseteq \mathcal{S}(X).$$

Since X is nonseparable, $I_X \notin \mathcal{S}(X)$, so $\mathcal{S}(X) \neq \mathcal{B}(X)$. Since $iJ \in \mathcal{S}(X) \setminus \mathcal{K}(X)$, we have

$$\mathcal{K}(X) \subsetneq \mathcal{S}(X) \subsetneq \mathcal{B}(X).$$

Consequently

$$\mathcal{S}(X)/\mathcal{K}(X)$$

is a non-zero proper closed two-sided ideal of $\mathcal{B}(X)/\mathcal{K}(X) = \mathcal{C}(X)$. Thus $\mathcal{C}(X)$ is not topologically simple. $\square$

8 The main theorem

Theorem 8.1. *There is a unital Banach algebra A which is not Banach-algebra isomorphic to $\mathcal{B}(X)/\mathcal{K}(X)$ for any Banach space X.*

Proof. Let $A = A_\lambda$ be the algebra constructed in Section 2. It is unital and topologically simple. By Lemma 3.1,

$$\operatorname{dens} A = \lambda = \beth_\omega.$$

Suppose, toward a contradiction, that there is a Banach-algebra isomorphism

$$\Phi : A \to \mathcal{C}(X)$$

for some Banach space X. Since Φ is a homeomorphism and A is topologically simple, $\mathcal{C}(X)$ is topologically simple. By Lemma 4.1,

$$\operatorname{dens} X \geq \lambda > 2^{\aleph_0}.$$

Choose distinct $\alpha_1, \alpha_2, \ldots \in \Gamma$, and put

$$\sigma_j = S_{\alpha_j} + M, \qquad \tau_j = T_{\alpha_j} + M$$

inside A. Then

$$\tau_i \sigma_j = \delta_{ij} 1_A,$$

and, by (1) and passage to the quotient,

$$\left\| \sum_j a_j \sigma_j \right\| \leq \max_j |a_j|$$

for every finitely supported scalar family (a_j). Also $\sup_j \|\tau_j\| \leq 1$.

Set

$$s_j = \Phi(\sigma_j), \qquad t_j = \Phi(\tau_j).$$

Then

$$t_i s_j = \delta_{ij} 1_{\mathcal{C}(X)},$$

$$\left\| \sum_j a_j s_j \right\| \leq \|\Phi\| \max_j |a_j|,$$

and

$$\sup_j \|t_j\| \leq \|\Phi\|.$$

Thus the hypotheses of Lemma 6.1 are satisfied. Hence X contains an isomorphic copy of c_0.

But $\operatorname{dens} X > 2^{\aleph_0}$, so X is nonseparable. Lemma 7.1 now implies that $\mathcal{C}(X)$ is not topologically simple. This contradicts the topological simplicity of A and the existence of the Banach-algebra isomorphism Φ.

Therefore no such Banach space X exists. □

9 Scope of the result

The proof separates three notions that are sometimes conflated.

(1) The algebra A_λ is topologically simple, because it is a quotient by a maximal proper closed ideal. The proof does not show that A_λ is algebraically simple; it may have dense non-closed ideals.

(2) The contradiction uses a closed ideal of $\mathcal{C}(X)$, namely $\mathcal{S}(X)/\mathcal{K}(X)$. A bounded Banach-algebra isomorphism pulls closed ideals back to closed ideals. A discontinuous algebraic isomorphism need not do so.

(3) The density argument and the estimate

$$\left\| \sum_j a_j \Phi(\sigma_j) \right\| \le \|\Phi\| \max_j |a_j|$$

both use boundedness of Φ. Therefore this proof establishes non-isomorphism in the Banach-algebra sense, not after forgetting the norms and allowing arbitrary discontinuous algebraic isomorphisms.

Thus Theorem 8.1 should be read exactly as a Banach-algebra statement: there is a unital Banach algebra not boundedly/topologically isomorphic to any Calkin algebra $\mathcal{B}(X)/\mathcal{K}(X)$.